

Shuheng Hu

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EDUCATION

Xi'an Jiaotong-Liverpool University, Jiangsu, China

Sept. 2023 – Jun. 2027 (Expected)

Bachelor of Science in Information and Computing Science, GPA: 3.87/4.0

Core courses: Artificial Intelligence, Machine Learning, Big Data Analysis, Bio Computation, Computer Graphics, Decision Computation and Language, Software Engineering, Human-Centric Computing, Discrete Mathematics and Statistics

RESEARCH INTERESTS

My research interests lie in **improving reasoning capabilities in multimodal and agentic AI**, with a focus on:

- Multimodal reasoning of LLMs and VLMs
- Autonomous agent frameworks with self-improvement
- Trustworthy and traceable systems over long horizons

PAPERS

- [1] Beyond the Code: Idea Tree-Driven Research Exploration in Automated Heuristic Design
Shuheng Hu, Cong Zhang, Ni Zhang, Zhiguang Cao. *Work in progress*
- [2] MechEvidence-Q: A Benchmark for Evidence-Grounded Verification of Quantitative Mechanobiology Claims
Shuheng Hu, Zimu Wang, Feng Xu, Haiyang Zhang. *Work in progress*
- [3] Multimodal Boolean Logic Culture Retrieval [\[paper\]](#)[\[code\]](#)
Jian Zhang, **Shuheng Hu**, Jiutian Chang, Junyi Guo, Huanda Lu, Qiufeng Wang, Fangyu Wu. **EMNLP 2026 Findings**
- [4] MuseBench: A Comprehensive Benchmark for Multimodal Cultural Understanding of Chinese Museum Artifacts
Jian Zhang, Junyi Guo, **Shuheng Hu**, Jiutian Chang, Qiufeng Wang, Fangyu Wu. *Work in progress*
- [5] Cross-Subject EEG Emotion Recognition via Adaptive Representation and Attention Fusion
Siyuan Gao, **Shuheng Hu**, Zhao Wang, Yumei Luo, Fangyu Wu. **BICS 2025 (oral)**
- [6] MoveMate: Supporting Video-Guided Workouts through Motion Visual Analysis and Multi-Device Guidance
Yihan Liu, Anqi Xie, **Shuheng Hu**, Yong Yue, Yu Liu. **MobileHCI 2026 (oral)** [\[paper\]](#)[\[code\]](#)

RESEARCH EXPERIENCE

Beyond the Code: Idea Tree-Driven Research Exploration in Automated Heuristic Design Jun. 2026 – Present

Advisor: [Zhiguang Cao](#) (Assistant Professor of Singapore Management University)

- Proposing an idea tree-driven research framework that records why each heuristic works as reusable hypotheses, a key blind spot of code-evolution methods
- Designing a quality-diversity idea generation strategy that balances exploiting proven heuristic directions with exploring structurally distinct design mechanisms, avoiding convergence to near-duplicate code
- Benchmarking against state-of-the-art LLM-based code evolution methods (FunSearch, MCTS-AHD) on combinatorial optimization tasks to test whether searching over ideas rather than code yields stronger and more interpretable heuristics

MechEvidence-Q: A Benchmark for Evidence-Grounded Verification of Quantitative Mechanobiology Claims Jun. 2026 – Present

Advisor: [Haiyang Zhang](#) (Assistant Professor of Xi'an Jiaotong-Liverpool University)

- Building a benchmark for verifying quantitative mechanobiology claims against their source evidence, motivated by the fact that such claims only hold under specific experimental conditions, which existing verification tasks rarely check
- Designing field-level accuracy metrics and hard-negative claims that differ from the evidence in only a single value or condition, so verifiers must check fine-grained details instead of relying on topical similarity
- Evaluating retrieval and LLM-based verification methods to find when claims stay evidence-grounded and when key experimental conditions or evidence boundaries are lost, toward more trustworthy AI for scientific literature

Multimodal Boolean Logic Culture Retrieval

Jan. 2026 – Aug. 2026

Advisor: [Fangyu Wu](#) (Assistant Professor of Xi'an Jiaotong-Liverpool University)

- Formulated a new task that evaluates cultural retrieval by logical validity rather than semantic overlap, motivated by the finding that current multimodal retrievers collapse Boolean queries into a single holistic embedding
- Proposed CLAIR, a training-free, plug-and-play framework that converts Boolean operators into fuzzy-logic scores over frozen embeddings, consistently improving both open- and closed-source models (up to +16.6 pts accuracy on VALSE)
- Built CulTi2.0, the first Boolean retrieval benchmark for cultural heritage (110K expert-verified queries), which reveals a substantial gap in logical reasoning: the best of eight mainstream embedding models achieves only 10.59 RSum

MuseBench: A Comprehensive Benchmark for Multimodal Cultural Understanding of Chinese Museum Artifacts

May 2025 – Present

Advisor: [Fangyu Wu](#) (Assistant Professor of Xi'an Jiaotong-Liverpool University)

- Identifying that existing museum benchmarks rely on web-scraped data and support only one task (VQA or retrieval), leaving VLMs' understanding of professional curatorial documentation unmeasured
- Building MuseBench, the first benchmark to unify Cultural VQA and Cultural Retrieval over official catalogs from 3 museums (29K artifacts, 128K multi-view images, 293K QA pairs), with a 98.16% human-audited QA pass rate
- Benchmarking 25 VLMs and embedding models, the best scored only 36.9% on VQA and $\leq 9.6\%$ R@1 on retrieval, with failures traced to professional terminology and structured metadata rather than reasoning

Cross-Subject EEG Emotion Recognition via Adaptive Representation and Attention Fusion

May 2025 – Nov. 2025

Advisor: [Fangyu Wu](#) (Assistant Professor of Xi'an Jiaotong-Liverpool University)

- Addressed the inter-subject distribution shift that limits EEG emotion recognition, where existing domain adaptation methods rely on rigid source-to-target alignment and lack a smooth transition between the two domains
- Developed CARAF-Net, which uses channel-wise cross-attention to build a learnable intermediate domain that bridges source and target subjects, and a dual-path extractor that captures spatial and temporal EEG features
- Reached state-of-the-art cross-subject accuracy: 89.05% on SEED and 76.30% on SEED-IV, beating PGCN by 2.61% and GMSS by 2.82%.

MoveMate: Supporting Video-Guided Workouts through Motion Visual Analysis and Multi-Device Guidance

Jun. 2024 – Jun. 2026

Advisor: [Yu Liu](#) (Assistant Professor of Xi'an Jiaotong-Liverpool University)

- Conducted a systematic analysis of online instructional workout videos and a contextual inquiry with users, identifying limited screen visibility as a key challenge in following video-guided workouts
- Designed MoveMate, a multi-device workout support system that recommends device placement before a workout and provides visual and auditory guidance during it, helping users follow movements across different body postures
- Evaluated MoveMate through user studies, showing that supporting users in following instructors under reduced screen visibility is a key design consideration and offering insights for future computer-assisted workout systems

SELECTED PROJECTS

Analyzing the Demand for Green Accountants | Core Developer

Jun. 2024 – Dec. 2025

Advisor: [Zhiying Huang](#) (Senior Teaching Fellow of Xi'an Jiaotong-Liverpool University), 2024 XJTLU Excellent Project

- Crawled, preprocessed, and maintained a database of 100K ESG job ads from major recruitment sites worldwide
- Analyzed the postings to find that non-accounting roles dominate sustainability hiring, green accounting roles cluster in markets with mature reporting frameworks, and employers increasingly value interdisciplinary expertise
- Designed an agentic workflow for industry ESG scenarios that asks targeted follow-up questions before concluding, decomposing and reflecting on the core steps of practitioners' daily workflows to distill them into a reliable AI workflow

SKILLS AND OTHERS

- **Language Ability:** English (Full professional proficiency), TOEFL 5.5 of 6 (110 of 120), Chinese (Native)
- **Programming language:** Python, Java, MySQL, C/C++, MATLAB, Rust, R, TypeScript, Kotlin
- **Online certification:** Google Cybersecurity Professional, DeepLearning.AI Retrieval Augmented Generation (RAG)
- **Framework & Tools:** PyTorch, TensorFlow, GitHub, Gradio, Postgres, Vue.js, Docker, Weaviate, HuggingFace, Neo4j
- **Interests:** Backpacking & Hiking, Meditation, Reading